

A Spectral Interpretation of Linear System Resolution via Normal Equations

Abstract

We present a structural reinterpretation of the classical normal-equation method for solving linear systems. Although the algebraic procedure is well known, we show that it can be understood as a boundary-induced spectral stabilization process, formally analogous to decoherence-driven diagonalization in quantum measurement. This interpretation provides a unified spectral–ontological reading of linear resolution, quantum collapse, and structural necessity.

1. Introduction

Given a linear system

$$Ax = y,$$

with $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$, and $y \in \mathbb{R}^n$, classical linear algebra introduces the normal equations

$$A^T Ax = A^T y$$

to obtain a symmetric positive definite system suitable for spectral decomposition.

While this method is standard in numerical analysis and least squares theory, its deeper structural interpretation has not been explicitly explored. In this work we show that the normal-equation transformation can be understood as a spectral reduction of incoherence induced by boundary projection.

2. Lemma 1: Symmetry and positivity

Let $A \in \mathbb{R}^{n \times n}$. Then the matrix $A^T A$ is symmetric:

$$(A^T A)^T = A^T A.$$

Furthermore, for any nonzero vector v ,

$$v^T A^T A v = \|Av\|^2 \geq 0.$$

If A is invertible, then $A^T A$ is strictly positive definite.

3. Lemma 2: Equivalence of solutions

If A is invertible, then the systems

$$Ax = y$$

and

$$A^T A x = A^T y$$

have the same unique solution.

Proof

Multiplying the first equation by A^T yields the second.

Conversely, since $A^T A$ is invertible, multiplying the second equation by $(A^T A)^{-1} A^T$ recovers the original solution. ■

4. Spectral decomposition

Since $A^T A$ is symmetric positive definite, by the spectral theorem there exists an orthogonal matrix Q such that

$$A^T A = Q D Q^T,$$

where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i > 0$.

Transforming variables:

$$z = Q^T x, \quad b = Q^T A^T y,$$

the system becomes

$$Dz = b,$$

with solution

$$z_i = \frac{b_i}{\lambda_i}.$$

Finally,

$$x = Qz.$$

Thus, the solution is expressed as a weighted combination of spectral modes.

5. Structural interpretation

This procedure shows that solving a linear system is equivalent to:

1. Projecting the problem into a symmetric boundary-induced operator $A^T A$.
2. Diagonalizing the induced operator.
3. Resolving the system along its spectral directions.
4. Reconstructing the solution as a superposition of spectral components.

The eigenvectors of $A^T A$ define privileged structural directions, while the eigenvalues determine their relative necessity.

6. Spectral–ontological interpretation

We interpret each eigenvector as a **structural delta**, representing a stable direction under boundary projection. The solution is not obtained by arbitrary inversion, but by spectral stabilization along these deltas.

Thus, resolution of a linear system is a process of **incoherence reduction into spectral deltas**.

7. Theorem (Spectral Resolution Theorem)

Theorem.

Let $A \in \mathbb{R}^{n \times n}$ be invertible. Then the solution of $Ax = y$ can be uniquely expressed as

$$x = \sum_{i=1}^n \frac{q_i^T A^T y}{\lambda_i} q_i,$$

where $\{q_i\}$ are the orthonormal eigenvectors of $A^T A$ and λ_i the corresponding eigenvalues.

This representation shows that any solution is a weighted superposition of structurally necessary spectral deltas.

8. Conceptual novelty

Although the normal-equation method is classical, its interpretation as a spectral collapse of incoherence induced by boundary projection has not been previously formalized.

This provides a conceptual bridge between:

- linear algebra,
 - spectral theory,
 - decoherence,
 - and quantum measurement structure.
-

9. Relation to quantum measurement

The diagonalization of $A^T A$ plays the same formal role as the diagonalization of a reduced density matrix in decoherence theory. In both cases, the system is resolved into stable spectral components selected by boundary interaction.

10. Conclusion

Solving a linear system is not merely an algebraic inversion procedure, but a structural spectral reduction process. The emergence of spectral deltas is mathematically inevitable and reflects a general principle of boundary-induced stabilization.

11. Closing sentence

Linear resolution, quantum decoherence, and structural collapse are different manifestations of the same spectral necessity principle.

12. Disclaimer

The algebraic method employed is classical; the novelty lies exclusively in its structural and ontological interpretation.